

Master Thesis Proposal

From Text to Cross-domain Causal Knowledge Graphs: An LLM-based Framework

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1. Abstract

This study proposes an LLM-based framework for cross-domain causal knowledge graph construction that directly produces structured causal triples from multi-domain text in a single generation pass. Existing methods either confine LLMs to specific pipeline stages, lack optimisation for causal semantics, or remain restricted to single domains, while graph construction approaches face limited ontology support, insufficient quality evaluation, and a lack of causal specificity. The proposed framework adopts GoLLIE’s annotation guideline technique to encode causal definitions, boundary rules, and output schema as first-class model inputs, constraining the LLM to output causal triples. A retrieval-augmented mechanism based on a causal pattern database extends zero-shot generation into few-shot generation to improve extraction quality. Generated triples undergo entity normalisation, Wikipedia linking, and conversion into RDFS/OWL representations aligned with a domain-independent causal ontology to construct the final knowledge graph. The framework is evaluated on three datasets (CauseNet [1], ADE [2], and Li et al. [3]) at both the triple and graph levels, with analysis of hallucination types and mitigation strategies.

Keywords: Causal Knowledge Graph, Large Language Models, Structured Generation, Annotation Guidelines, Retrieval-Augmented Generation

2. Introduction

Causality is one of the most fundamental cognitive structures through which humans understand the world. From adverse drug reactions in medicine to the attribution of market fluctuations in finance and the analysis of policy impacts in news, identifying and organising causal knowledge has broad application value for decision support, knowledge discovery, and intelligent reasoning. Extracting causal relations from text and organising them into structured causal knowledge graphs has become an important research direction at the intersection of natural language processing and knowledge engineering.

However, reliably constructing causal knowledge graphs from unstructured text remains a challenging task. Traditional approaches rely on pattern matching, feature engineering, and multi-stage pipelines [3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19], which can be effective for explicit causal expressions but are heavily dependent on manual annotation and domain-specific design, with limited cross-domain transferability. The emergence of large language models (LLMs) has opened new possibilities: some work employs LLMs for causal detection, classification, or retrieval augmentation [20, 21, 22, 23, 24, 25], while others explore end-to-end structured generation [26, 27, 28, 29, 30, 31, 32]. Yet existing methods either confine LLMs to specific pipeline stages, achieve end-to-end generation without optimising for the semantic

specificity of causal relations, or remain restricted to single domains. On the graph construction side, resource-based methods [33, 34, 35, 36, 37, 38, 39, 40, 41, 1, 42, 43, 44] remain dependent on domain-specific expert resources with limited generalisability, while generative approaches [45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57] still follow sequential construction processes and lack systematic evaluation of causal knowledge quality. Overall, causal extraction and graph construction remain insufficiently integrated, and existing studies have not yet provided a reliable cross-domain framework that unifies causal triple generation with graph construction in a coherent workflow.

The present moment is well suited for this line of research. GoLLIE [29] has demonstrated that constraining LLM structured output through annotation guidelines can achieve low-hallucination, cross-domain zero-shot information extraction, providing a viable technical path for generative causal triple extraction. The emergence of domain-independent causal ontologies, such as the causal Ontology Design Pattern proposed by Jaimini et al. [58], enables generated outputs to be aligned to a unified semantic framework, offering ontology-level support for cross-domain graph construction. The maturation of retrieval-augmented generation (RAG) techniques [20, 21, 22] further provides complementary means to improve generation quality and cross-domain robustness. The concurrent readiness of these technical conditions makes it feasible to build a unified cross-domain generative framework from text to causal knowledge graph.

Against this background, this study proposes an LLM-based framework that unifies cross-domain causal triple extraction and causal knowledge graph construction within a single generative workflow. By encoding causal definitions and output schemas as guideline-constrained inputs, augmenting generation with retrieved causal patterns, and aligning outputs with a domain-independent causal ontology, the framework concentrates the core extraction workload in one generation pass while preserving ontology-aware graph organisation. This work is addressed to researchers in knowledge graph construction, causal reasoning, and LLM-based information extraction, and is intended to be transferable to application scenarios such as healthcare, finance, and news analysis where reliable causal knowledge is required. The remainder of this proposal first introduces the necessary preliminaries (Section 3) and reviews related work (Section 4), then formulates the research problem and questions (Section 5), and finally presents the proposed methodology and evaluation plan (Section 6).

3. Preliminaries

3.1. Causality and Causal Relations

3.1.1. Definition of Causality

Causality refers to a relationship in which the occurrence of an event, state, or condition brings about the emergence or change of another event, state, or condition [59]. For example, “smoking

causes lung disease” describes a typical causal relation, where the former (smoking) gives rise to the latter (lung disease) through an underlying mechanism. Formally, causality has two key properties: directionality, meaning that the cause and the effect are not interchangeable; and dependency, meaning that the effect depends on the presence or influence of the cause rather than merely co-occurring with it [59]. In this study, causality is understood as a directed and explanatory dependency that explicitly captures a “cause–effect” relationship, rather than a general notion of association or co-occurrence.

3.1.2. Causality and Correlation

Causality should be distinguished from correlation. Correlation describes a statistical co-variation between two variables, where changes in one variable are associated with changes in another. However, such co-variation does not imply that one causes the other [59]. For instance, ice cream sales and the number of drowning incidents may be positively correlated, yet neither causes the other; instead, both are influenced by a common external factor such as temperature.

Causality goes beyond correlation by requiring both a directional relationship and an explanatory dependency, where the effect can be accounted for in terms of the cause [59]. In contrast, correlation alone does not specify directionality and may arise from coincidental patterns or shared underlying factors. This distinction is illustrated in Figure 1.

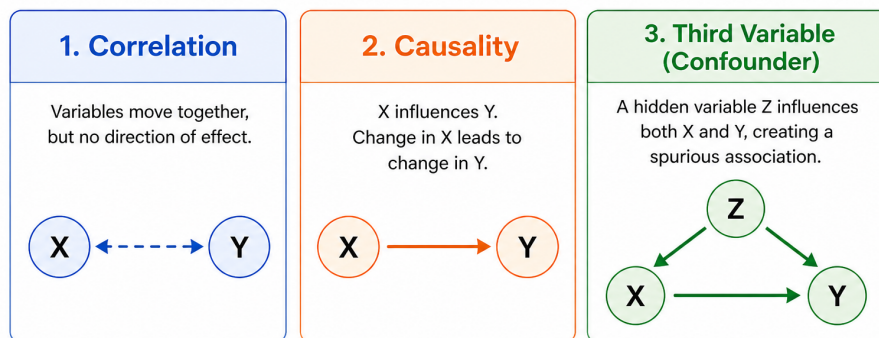


Figure 1: Illustration of the distinction between correlation and causality. A third variable (confounder) may influence both variables, leading to a spurious association.

In natural language processing tasks, this distinction is particularly important. The frequent co-occurrence of two concepts in text does not constitute evidence of causation. Therefore, causal relation extraction cannot be reduced to identifying associative patterns, but must instead capture underlying semantic directionality and dependency [60, 61].

3.1.3. Types of Causal Relations

Causal relations in text do not always appear in a uniform form. Depending on their linguistic expression and structural complexity, they can be categorized along multiple dimensions. Two particularly important distinctions are between explicit and implicit causality, and between direct and mediated causation [60, 61]. These distinctions have direct implications for downstream tasks such as causal relation extraction, structured representation, and knowledge graph construction.

Explicit and Implicit Causality Explicit causality refers to cases where causal relations are directly indicated by linguistic markers, such as “because,” “due to,” “result in,” and “lead to.” In such cases, the presence and direction of causality can be identified from surface-level cues [60, 61]. In contrast, implicit causality refers to relations that are not signaled by explicit markers but are instead inferred from context, semantics, or domain knowledge [60, 13]. For example, in the sentence “The company significantly reduced its R&D budget. The following year, its market share declined,” no explicit causal connective is used, yet a causal relation can be inferred. In real-world texts, a substantial portion of causal information is expressed implicitly, which implies that causal identification cannot rely solely on keyword matching but requires deeper semantic understanding [60, 61, 13].

Direct and Mediated Causation Direct causation refers to cases where the cause directly leads to the effect without intermediate steps. Mediated causation, by contrast, involves one or more intermediate variables (mediators) through which the cause influences the effect [58]. For instance, “smoking causes lung cancer” can be seen as a simplified expression of a more detailed causal mechanism, where smoking induces chronic inflammation, which in turn increases the risk of lung cancer. Similarly, in financial contexts, policy changes may influence asset prices indirectly through their impact on market sentiment. Distinguishing between direct and mediated causation is important for structured knowledge representation: representing only flat cause–effect pairs may overlook intermediate mechanisms, whereas explicitly modeling mediators allows for a more faithful representation of causal pathways [58].

3.1.4. Textual Representations of Causal Relations

The same causal relation can be expressed in text at different levels of granularity and in different structural forms. Selecting an appropriate representation is therefore essential for causal relation extraction and downstream processing [60, 61]. As illustrated in Figure 2, existing approaches typically adopt three main forms of representation: word pairs, span pairs, and causal triples.

Word Pairs Word pairs represent causality using two concept words or short phrases, such as smoking → cancer or investment → bankruptcy [61]. This representation is simple and suitable

Textual Representations of the Same Causal Relation

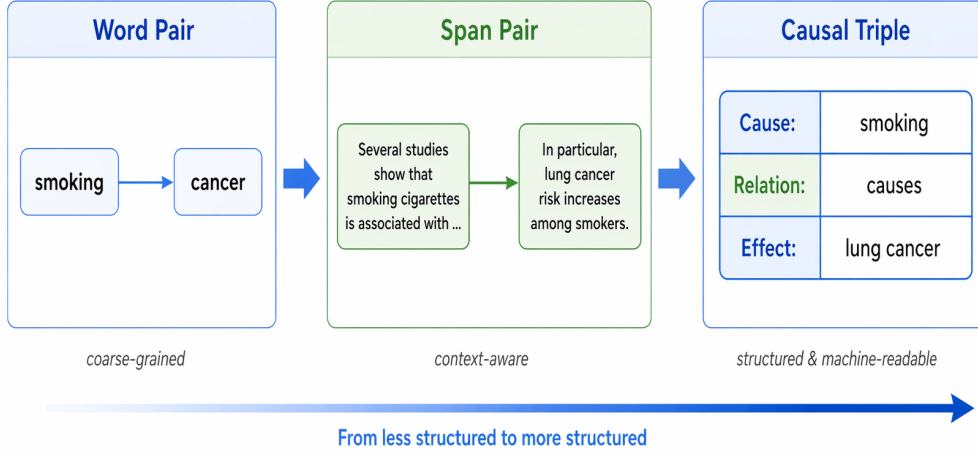


Figure 2: Illustration of different textual representations of the same causal relation. The representations progress from less structured (word pairs) to more structured and machine-readable forms (causal triples).

for large-scale aggregation and statistical analysis. However, it lacks contextual information and often fails to capture semantic nuances, modifiers, and temporal conditions present in the original text.

Span Pairs Span pairs represent causal relations using two text spans corresponding to the cause and the effect, for example, “Because of their wrong investment” → “they went bankrupt” [13, 60]. Compared to word pairs, span pairs preserve richer semantic information and remain closer to the original linguistic expression. As a result, they are more suitable for text-based extraction tasks. However, this representation also introduces additional challenges, particularly in accurately identifying the boundaries of cause and effect spans.

Causal Triples Causal triples provide a more structured representation by organizing causal knowledge into a triplet form (cause, relation, effect) [60, 61]. This format combines semantic clarity with structural consistency: it explicitly captures the cause–effect relation while also allowing different types of causal relations to be distinguished through relation labels. Moreover, its structured nature makes it well suited for downstream processing, such as automated parsing, knowledge storage, and knowledge graph construction. In this study, causal triples are adopted as the primary representation of causal knowledge.

3.1.5. Application Domains

The identification and organization of causal knowledge have practical value across multiple domains [61]. In the medical domain, extracting causal relations from clinical guidelines and biomedical literature, such as those between drugs and adverse effects, can support diagnosis and pharmacovigilance. In the financial domain, identifying causal mechanisms among market events can facilitate risk attribution, investment analysis, and policy evaluation. In news analysis, constructing causal networks of events can support tasks such as trend tracking, event forecasting, and policy impact assessment. Across these domains, common challenges include large volumes of textual data, diverse expressions of causality, and a strong requirement for the reliability of extracted causal knowledge.

3.2. Knowledge Graphs

3.2.1. Basic Definitions and Structure

A knowledge graph is a graph-based method for representing knowledge, in which nodes denote real-world objects, concepts, or events, and edges denote semantic relations between them [62]. The core representational unit of a knowledge graph is the triple, usually written in the form of (head, relation, tail), where head and tail are nodes and relation is the directed edge connecting them [62]. For example, (smoking, causes, lung cancer) constitutes a triple expressing a causal relation. The advantage of graph structures lies in their ability to naturally represent complex relations among entities, while also supporting tasks such as path-based reasoning, pattern discovery, and knowledge querying [62]. Once structured information extracted from text has been normalized, it can be organized into the form of a knowledge graph, enabling systematic storage and reuse of knowledge. To ensure that knowledge can be represented, shared, and extended consistently across systems, knowledge graphs typically rely on standardized semantic representation frameworks.

3.2.2. Knowledge Representation Languages

Constructing a knowledge graph requires not only defining its topology, but also adopting unified representation standards so that knowledge can be processed by machines, shared across systems, and extended in later stages. The most widely used standards for this purpose are RDF and its higher-level extensions, RDFS and OWL [63, 64, 65].

RDF The Resource Description Framework (RDF) is a knowledge representation standard developed by the W3C. Its basic idea is to describe resources and their relations using triples of the form subject, predicate, and object [63]. Each triple expresses one atomic fact. For example, “smoking” (subject) “causes” (predicate) “lung cancer” (object) represents a simple

causal statement. RDF provides a unified and machine-readable foundation for structured knowledge, allowing knowledge from different sources to be integrated and queried within a common framework [63].

RDFS and OWL RDFS (RDF Schema) and OWL (Web Ontology Language) extend RDF by providing richer semantic expressiveness [64, 65]. If RDF mainly answers the question of “what facts are present,” RDFS and OWL further address “what types these facts belong to” and “what rules they follow.” More specifically, RDFS introduces concepts such as classes, subclass relations, and properties, allowing entities and relations to be organized hierarchically [64]. OWL builds on this by offering more expressive logical constructs, including property constraints, equivalence and disjointness between classes, and cardinality restrictions, thereby supporting more fine-grained semantic modeling and automated reasoning [65]. In practical knowledge graph construction, RDFS and OWL are commonly used to define the schema layer of a graph, specifying the type system and constraints that nodes and edges should follow. Built on these standards, knowledge graphs often rely on ontologies to further formalize their conceptual structure and semantic constraints.

3.2.3. Ontology and Ontology Design Patterns

In the context of this study, an ontology refers to a formal specification of the concepts, relations, and organizational structure within a knowledge domain. It is not used here in the philosophical sense, but as a semantic modeling tool that defines what types of nodes and relations exist in a knowledge graph and how they are organized [66]. The role of an ontology is to provide a consistent semantic framework for knowledge representation, so that data from different sources can be integrated under a shared conceptual system, while also supporting reuse and extension [66].

T-Box and A-Box An ontology typically contains two levels of content. The T-Box (terminological box) represents the schema level, where concept classes, relation types, and their hierarchical structure are defined. For example, in a causal knowledge graph, the T-Box may define concept classes such as Cause, Effect, and Mediator, as well as relation types such as causes and causesWith. The A-Box (assertional box) represents the instance level, where concrete entities and facts are recorded. For example, “smoking causes lung cancer” is a specific assertion at the A-Box level. The separation between T-Box and A-Box allows the same conceptual framework to be reused across different domains of data, thereby improving the reusability of the ontology [66].

Ontology Design Patterns Ontology Design Patterns (ODPs) are reusable semantic modeling templates designed to address recurring structural problems in knowledge modeling [66]. Com-

pared with building a complete domain ontology from scratch, ODPs provide a lighter-weight solution by encapsulating common modeling structures that can be embedded into different domain ontologies [66]. For example, an ODP designed for causal relations may define a standard organization of causes, effects, and mediators, without requiring each application domain to redesign the causal schema from the beginning. The value of ODPs lies in reducing the complexity of ontology construction while preserving semantic consistency, allowing the same modeling pattern to be applied across domains.

3.2.4. Causal Knowledge Graphs

A Causal Knowledge Graph is a knowledge graph specifically designed to represent causal knowledge, in which nodes and edges are organized around causes, effects, and related concepts. Compared with general-purpose knowledge graphs, causal knowledge graphs are more specialized in their semantic goals [23, 1].

Differences from General-purpose Knowledge Graphs The differences between causal knowledge graphs and general-purpose knowledge graphs can be understood along three main dimensions. First, in terms of relation types, general-purpose knowledge graphs typically include a wide range of semantic relations, such as founded by, located in, or produced by, whereas causal knowledge graphs focus on causal relations such as causes, affects, and mediated causal connections [23, 1]. Second, in terms of node types, general-purpose knowledge graphs are often centered on named entities such as people, places, and organizations, while causal knowledge graphs more often represent events, states, or abstract concepts, such as smoking, heavy rain, market sentiment, or price fluctuation [23, 1]. These nodes are generally more difficult to identify and normalize than standard named entities. Third, in terms of semantic structure, causal knowledge graphs place greater emphasis on directionality and mechanism. Rather than merely recording factual connections, they aim to represent how one event or state leads to another, and may include intermediate links that capture possible causal pathways [23]. Figure 3 illustrates this contrast: the left side shows a general-purpose knowledge graph centered on entities and general semantic relations, while the right side shows a causal knowledge graph centered on event- or state-based nodes, directed causal relations, and possible mediating paths.

Challenges in Causal Knowledge Graph Construction Constructing a causal knowledge graph involves several distinctive challenges. The first is node redundancy. The same causal concept may appear in different texts under different surface forms, such as “economic recession,” “economic slowdown,” and “economic contraction.” Without normalization, the graph may contain multiple nodes referring to the same concept [23, 1]. The second is semantic ambiguity. The same expression may refer to different concepts in different contexts, and the causal direction between the same pair of concepts may even be inconsistent across sources. If such ambiguities

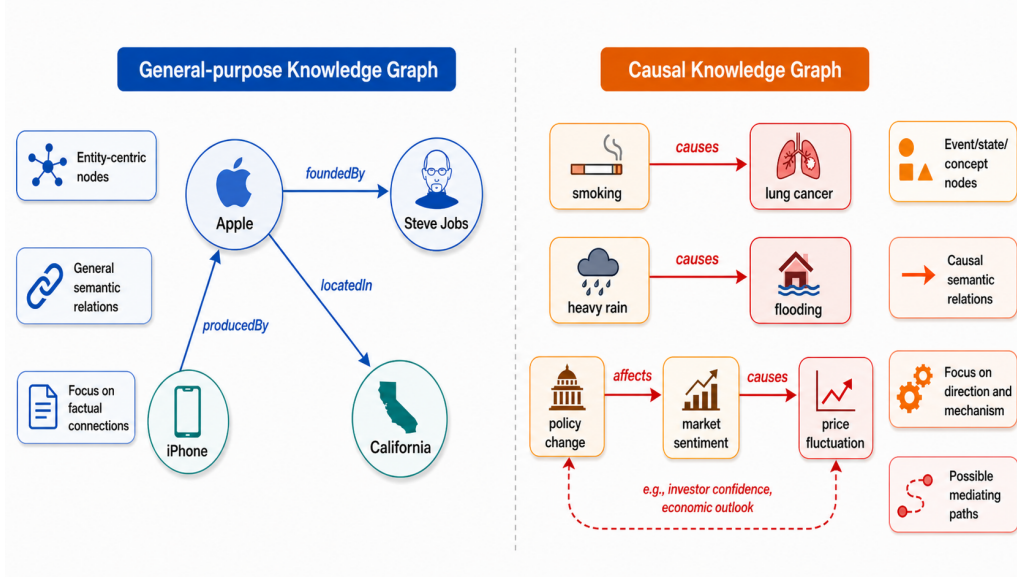


Figure 3: Comparison between a general-purpose knowledge graph and a causal knowledge graph. The former mainly represents entities and general semantic relations, whereas the latter focuses on event- or state-centered nodes, directed causal relations, and possible mediating paths.

are not resolved, they can undermine the consistency of the graph [23]. The third is chain coherence. Locally correct causal triples do not necessarily combine into semantically coherent causal chains. Two independently valid causal pairs may produce semantic discontinuity or logical inconsistency when linked together. Therefore, the construction of a causal knowledge graph must address not only the correctness of individual causal relations, but also the overall plausibility of multi-hop causal paths [23]. These challenges suggest that building causal knowledge graphs depends not only on relation extraction itself, but also on later stages such as node normalization and external knowledge alignment.

3.2.5. Entity Linking and Interoperability

Entity linking refers to the process of mapping nodes in a knowledge graph to standardized entity or concept identifiers in external knowledge bases [67]. For example, a node such as “lung cancer” in a graph may be linked to its corresponding entry in Wikipedia or Wikidata. The purpose of entity linking is to reduce ambiguity in node expressions and provide each concept with a unique standard identifier, thereby improving the interpretability and precision of the graph [67].

On this basis, entity linking also supports interoperability, that is, the ability of different knowledge resources to connect, share, and reuse information [67]. When nodes in multiple graphs are linked to the same external identifier, correspondences between them can be estab-

lished, making cross-graph integration and joint querying possible. Entity linking is particularly important for causal knowledge graphs because their nodes are often events or abstract concepts rather than standard named entities. As a result, the diversity of textual expressions is usually much greater than in general-purpose knowledge graphs. Without entity linking, redundancy and ambiguity at the node level become even more severe.

3.3. Large Language Models

3.3.1. Definition and Core Characteristics

Large Language Models (LLMs) are language models based on the Transformer architecture and pretrained on large-scale text corpora, with strong capabilities in natural language understanding and generation [68]. The term “large” mainly refers to two aspects: model size, which often reaches billions of parameters or more, and training data scale, which typically covers massive cross-domain text corpora. The core characteristics of LLMs can be summarized as follows: they can take natural language as input and model its semantic structure; they can generate coherent and fluent text conditioned on context; and they can exhibit a certain degree of generalization across multiple downstream tasks even without explicit task-specific training.

In recent years, the LLM ecosystem has become increasingly diverse. Proprietary models, represented by OpenAI’s GPT series [69], Anthropic’s Claude series [70], and Google’s Gemini series [71], generally reflect the development of general-purpose capabilities, multimodal processing, tool use, and production-level service support. For example, OpenAI’s official documentation describes GPT-4.1 as a model with strong instruction-following and tool-calling capabilities, Anthropic’s Claude series emphasizes integrated performance in complex language tasks, analytical reasoning, and safety alignment, and Google’s Gemini documentation highlights its capabilities in long-context processing, multimodal input, and structured output [70, 69, 71]. By contrast, open or open-weight models, represented by Meta’s Llama series [72], Alibaba’s Qwen series [73], and DeepSeek [74], demonstrate greater flexibility for fine-tuning, private deployment, research reproducibility, and engineering adaptation. For example, Meta describes Llama as an open model that can be fine-tuned and deployed, Qwen provides both commercial and open-source model families, and DeepSeek places particular emphasis on performance in complex reasoning tasks, especially on chain-of-thought reasoning [72, 73, 74]. Together, these models have pushed large language models beyond general text generation tools toward platforms for structured knowledge processing, and have provided a richer technical foundation for tasks such as information extraction, knowledge generation, and knowledge graph construction.

3.3.2. Training and Adaptation Paradigms

The capabilities of LLMs originate from large-scale pre-training, while their performance on specific tasks further depends on subsequent adaptation strategies. Common adaptation paradigms

include fine-tuning (and parameter-efficient adaptation) and prompt engineering; together with pre-training, they form a complete technical chain from general language capability to task-oriented behavior. These paradigms differ in computational cost, flexibility, and dependence on annotated data [75, 76].

Pre-training Pre-training is the foundational stage through which LLMs acquire general language capabilities. At this stage, the model learns statistical regularities of language, semantic structure, and distributed representations of world knowledge from large-scale unlabeled text data through self-supervised learning objectives, such as next-token prediction. The pretrained model is not optimized for any specific downstream task; rather, it develops a general basis for language understanding and generation, which serves as the starting point for subsequent task adaptation. The quality and scale of pre-training largely determine the upper bound of the model’s general capabilities.

Fine-tuning and Parameter-efficient Adaptation Fine-tuning refers to further updating the parameters of a pretrained model using annotated data from a specific task or domain, so that the model becomes better suited to the target application scenario. Fine-tuned models often achieve substantial performance gains on the target task, but this usually requires sufficient labeled data and corresponding computational resources. To reduce the cost of fine-tuning, parameter-efficient adaptation methods have been proposed. A representative example is LoRA (Low-Rank Adaptation), which updates only a small trainable subspace of the model parameters while leaving most pretrained parameters unchanged, thereby enabling task adaptation with much lower computational cost [75]. Such methods are particularly valuable in scenarios where models need to be deployed in specialized domains, such as medical or financial text processing.

Prompt Engineering Prompt engineering is a task adaptation strategy that does not modify model parameters, but instead guides model behavior through the design of input prompts. Its core idea is to organize task instructions, constraints, and optional examples as part of the model input, so that the model follows the intended output pattern during generation. Depending on whether examples are included in the input, prompt engineering can be further divided into zero-shot and few-shot settings. In the zero-shot setting, only task instructions are provided, and the model relies entirely on its internal capabilities to complete the task. In the few-shot setting, a small number of completed input–output examples are additionally provided in the prompt, allowing the model to infer the task pattern by analogy [76]. Prompt engineering is attractive because it is highly flexible and does not require additional training, and is therefore widely used in rapid prototyping and cross-task applications.

3.3.3. Generative Information Extraction

The core idea of Generative Information Extraction is to reformulate traditional information extraction tasks as text generation tasks: given an input text, the model directly generates structured extraction results instead of deriving them step by step through discriminative methods such as classification or sequence labeling [77]. Information extraction itself covers multiple subtasks, including entity recognition, relation extraction, and event extraction. The key characteristic of the generative paradigm is that it replaces modular processing with a unified generation framework, allowing the model to produce the final structured output in a single process [31, 77].

From Pipeline-based Extraction to Generation Traditional information extraction systems usually adopt a pipeline architecture, in which the overall task is decomposed into multiple sequential stages. For example, a system may first perform named entity recognition, then relation extraction, and finally entity linking. The advantage of such a design is that each module has a clear objective and can be trained and evaluated independently. However, its main limitation is error propagation: mistakes made by upstream modules inevitably affect the quality of downstream inputs, causing errors to accumulate over the pipeline. Generative approaches attempt to avoid this problem by using a single model to generate the final structured output directly from the source text, thereby unifying multiple implicit subtasks within one generation process. This change is not simply a matter of “turning classification into generation,” but rather a shift in task organization from a multi-stage pipeline to a unified model centered on the final output [31, 77].

Single-pass Structured Output Single-pass structured output refers to the direct generation of parseable structured results in a single generation step, such as JSON, tables, or code-like objects. Compared with free-form text, structured output has clearer field constraints and format specifications, allowing downstream systems to read and process it directly without additional parsing steps [31]. This form is particularly suitable for scenarios in which extraction results are further used for knowledge storage, knowledge graph construction, or automated reasoning. For example, for the sentence “Smoking causes lung cancer,” a model may directly generate a structured object such as {cause: smoking, relation: causes, effect: lung cancer}, instead of separately outputting entities, relation labels, and post-processed results. Likewise, recent versions of both proprietary and open models have begun to support structured outputs or tool-based interaction, which further enhances the practical feasibility of generative extraction [69, 72].

In structured generation settings, output quality depends not only on whether the content is correct, but also on whether the format is consistent, the fields are complete, and the result is parseable. For this reason, generative information extraction usually requires explicit task

instructions, annotation guidelines, or output schemas to constrain model behavior. More specifically, such constraints typically cover at least three aspects: first, the definition of the target extraction object, that is, what should and should not be extracted; second, boundary and scope rules, which specify what words or spans should be included in each output field; and third, structural output requirements, which determine how the result should be organized in terms of fields, hierarchy, or data types. For example, in causal relation extraction, the system must not only tell the model that it should identify cause, relation, and effect, but also specify what expressions count as causal relations, how the boundaries of cause and effect spans should be determined, and what structured format the final output should follow. In this way, the model no longer freely generates a plausible piece of text, but instead produces a structured result under explicit task constraints [29]. As structured output capabilities continue to mature in general-purpose models, this form of schema- and guideline-based control has become increasingly practical [29, 69].

3.3.4. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a technical paradigm that combines external knowledge retrieval with language model generation [78]. Its basic workflow is as follows: before generation, the system first retrieves relevant information from an external knowledge base, vector database, or document collection based on the input, and then injects the retrieved content into the model’s input context as additional evidence for generation [78]. The central idea of RAG is to combine the model’s parametric knowledge with externally retrieved knowledge obtained at runtime, so that the model does not have to rely solely on what it has memorized during pre-training, but can instead consult information that is more recent, more relevant, or more reliable [78]. In recent years, this paradigm has developed into a relatively mature engineering ecosystem. For example, LangChain organizes retrieval as composable chain-based modules that support joint retrieval and generation, while LlamaIndex emphasizes document indexing, retrieval, and knowledge interface construction, making it easier to connect external data to LLM applications [79, 80]. As a result, RAG is no longer merely a research concept, but has become a commonly used framework for practical system development.

Retrieval Mechanisms The effectiveness of RAG largely depends on whether the retrieval stage can identify genuinely relevant information. Common retrieval methods can be divided into three broad categories. Dense retrieval encodes text into continuous vector representations and retrieves candidates based on semantic similarity, which is effective for capturing deeper semantic relations. Sparse or feature-based retrieval, by contrast, relies on keywords, structured features, or rules, and is often more precise for matching specific surface patterns. Hybrid retrieval combines the two approaches in an attempt to exploit the complementary strengths of semantic similarity and surface-level features. The choice of retrieval strategy depends on the

nature of the task. For tasks with complex semantics and diverse expressions, dense retrieval is often more effective; for tasks with explicit surface cues, feature-based retrieval may be more precise.

Role in Information Extraction In information extraction tasks, the main role of RAG is to retrieve similar completed examples and provide the model with task-relevant few-shot demonstrations or background evidence, thereby improving the accuracy and consistency of generation [78]. More specifically, when the model receives a new extraction input, RAG can retrieve cases from a pre-built instance repository that are structurally or semantically similar to the current input and include them as reference examples in the prompt. In this sense, the mechanism dynamically extends zero-shot generation into retrieval-supported few-shot generation, allowing the model to better capture the target task pattern through analogy with prior examples [78]. In addition, retrieved external information can provide stronger grounding for generation and reduce the risk of producing unreliable outputs in the absence of sufficient evidence. For example, in causal relation extraction, the system may prioritize examples with similar causal structures or trigger patterns and use them as references for the current input, thereby improving the stability of the generated results.

3.3.5. Hallucination and Faithfulness in LLM-based Generation

Hallucination refers to cases in which a language model generates content that is unsupported by the input text or inconsistent with objective facts [81, 82]. In information extraction and knowledge generation scenarios, hallucination may take several forms, including generating entities or relations that do not appear in the source text, incorrectly identifying entity boundaries, or constructing facts that appear plausible but lack textual support [81, 82]. For example, in causal relation extraction, a model may generate a causal relation when the source text only expresses correlation; it may introduce a mediator even though no such mechanism is explicitly mentioned in the text; or it may truncate “long-term smoking” to “smoking,” thereby omitting a crucial modifier and distorting the intended meaning [81, 82]. These cases show that hallucination does not necessarily involve entirely fabricated content; it may also manifest as partial distortion, boundary errors, or misclassification of relation types.

Faithfulness refers to the degree to which generated output remains grounded in the source text, that is, whether the output is genuinely supported by the input rather than supplemented or invented by the model itself [81, 82]. In causal knowledge generation, the requirement for faithfulness is especially strict: every element of a generated causal triple, namely the cause, relation, and effect, should be explicitly supported by the source text. If a model generates a causal relation that is not mentioned in the text, then even if the relation appears reasonable from the perspective of commonsense knowledge, it should still be regarded as a hallucinated output because it goes beyond the available textual evidence. Therefore, in LLM-based structured

knowledge generation, faithfulness is not only an important criterion for evaluating generation quality, but also a prerequisite for ensuring the verifiability and usability of the output.

4. Literature Review

4.1. Causality Extraction

4.1.1. Traditional Approaches

Pattern-Based Methods Early causal relation extraction relied primarily on identifying explicit causality cues in text. Researchers constructed handcrafted extraction rules based on syntactic, lexical, and grammatical patterns derived from causative verbs, causal connectives, and resultative constructions [13, 17, 18, 19]. Khoo et al. [19] first parsed sentences into syntactic parse trees and then defined graphical extraction patterns over these tree structures to locate cause–effect pairs from biomedical abstracts. These patterns exploited the structural relationships among syntactic constituents to identify causal expressions, achieving high precision within the coverage of the defined rules. Girju [18] combined causal verb templates with WordNet-based semantic constraints to build a causal question answering system. Rather than relying solely on surface-level verb matching, this method used the semantic hierarchies provided by WordNet to filter candidate causal pairs at the semantic level, thereby improving matching accuracy. However, the construction cost of these methods was high, and once a pattern set was established it was difficult to extend automatically to new domains or text types.

To reduce dependence on static pattern sets, subsequent work explored automated and semi-automated pattern acquisition. Chang and Choi [17] proposed a bootstrapped cue-phrase learning framework that started from a small set of seed cues and iteratively discovered new causal indicators from unannotated corpora using probabilistic confidence scores, progressively expanding the coverage of causal patterns. Beamer and Girju [15] explored unsupervised causal prediction from a statistical perspective, computing a causal potential score from co-occurrence statistics over temporally adjacent event bigrams (i.e., pairs of events that appear in close temporal sequence) in narrative corpora to estimate whether two events are likely to be causally related. Ittoo and Bouma [13] further extended the extraction scope to implicit causality. They systematically collected multiple types of causal patterns from Wikipedia, including resultative verbs, inseparable cause–result constructions, and non-verbal markers, and transferred these patterns to domain-specific corporate texts for causal extraction, contributing a taxonomy of implicit causal types. Despite these efforts to broaden the acquisition and coverage of causal patterns, pattern-based methods remained fundamentally constrained by the coverage of the patterns themselves: causal expressions that fell outside the pattern set could not be identified, and this limitation was particularly pronounced in implicit causality modelling and cross-domain transfer.

Shallow Machine Learning Methods Shallow machine learning methods refer to approaches that employ traditional supervised classifiers (such as logistic regression and support vector machines) and structured inference techniques for causal relation extraction. Unlike pattern-based methods, these approaches do not rely on handcrafted fixed rules but instead learn decision boundaries automatically from annotated data, using feature engineering to encode lexical, syntactic, and semantic information from text as input features for the classifier, thereby enabling more flexible causal identification [83].

Blanco et al. [16] proposed a two-stage pipeline in which the first stage located candidate causal sentences via syntactic templates, and the second stage applied a classifier with lexical, syntactic, and semantic features to filter out false positives. This method operated under the assumption that the text contains abundant explicit causal markers (i.e., marker-rich text refers to text containing clear causal connectives or trigger words such as “because” and “due to”), and therefore lacked the ability to identify implicit causal relations where such explicit markers are absent. Do et al. [14] combined distributional cause–effect association scores with discourse connective evidence and reconciled local predictions through Integer Linear Programming (ILP) [84] for global consistency, improving cross-sentence causal identification. The approach first independently estimated the likelihood of each candidate causal pair at the local level, then applied global constraints via ILP (such as transitivity and mutual exclusivity of causal relations) to ensure document-level consistency in the final output. However, this architecture depended on the sequential passing of information through multiple upstream components, making it susceptible to cascading errors.

Mirza and Tonelli [12] developed CATENA, which employed a hybrid sieve-based architecture. In this design, multiple filters (sieves), each using different features or rules, are arranged in a priority-ordered cascade: candidates that cannot be resolved by one sieve are passed to the next for further processing. The core design of CATENA lay in jointly extracting temporal and causal relations, leveraging the constraint that causality implies temporal precedence (i.e., a cause typically precedes its effect in time). This cross-task constraint was encoded into the sieve pipeline, demonstrating the complementary benefits of joint temporal–causal modelling. However, the system remained confined to explicitly annotated causal links and offered limited coverage of implicit causality. Gao et al. [7] combined logistic regression classifiers with document-level ILP constraints that encoded structural assumptions such as main-event centrality (i.e., the central events in a document are more likely to participate in causal relations) and event coreference (i.e., different mentions referring to the same event should receive consistent causal judgements), in order to recover sparse cross-sentence causal links. This work represented a mature instantiation of the shallow machine learning paradigm for document-level causal reasoning.

Overall, shallow machine learning methods advanced causal extraction from pattern engineering to feature engineering and multi-stage pipelines. While learned decision boundaries

and global inference mechanisms brought notable improvements, cascading error propagation and fragile cross-domain performance remained core bottlenecks of this paradigm.

Deep Learning Methods Deep learning freed causal extraction from laborious feature engineering through automatic representation learning. Unlike shallow machine learning methods that rely on manually designed lexical, syntactic, and semantic features, deep learning models can automatically learn effective feature representations directly from raw text, reducing dependence on domain expert knowledge.

In the broader field of relation extraction, generative and unified architectures had already achieved notable progress. REBEL [85] reframed end-to-end relation extraction as autoregressive seq2seq generation using BART, with the model directly outputting linearized relation triplets (i.e., triplets that originally have a graph structure are converted into a flat text sequence using predefined special tokens, making them suitable as generation targets for a seq2seq model). UniRel [86] verbalized relation labels into natural language expressions and jointly encoded them with sentence tokens, modelling entity–entity and entity–relation dependencies within a unified interaction map, thereby avoiding the separation of entity recognition and relation classification into independent stages. ReLiK [87] adopted a retrieve-and-read paradigm to perform entity linking and relation extraction in a single forward pass: a retriever first obtains relevant entities and relations from a candidate set, and a reader then determines their validity within the context of the input text. However, when applied to causal relations, these general-purpose techniques are often limited by the semantic complexity inherent in causality [10, 11]. Causal relations involve not only semantic associations between entities but also require the model to understand directionality, implicit expressions, and cross-sentence reasoning, which makes general-purpose methods less effective in causal scenarios.

Within specialized causal extraction research, early deep models were built on convolutional neural networks (CNNs) [88] and recurrent neural networks (RNNs) [89]. Kruengkrai et al. [10] introduced a multi-column CNN architecture that assigned separate convolutional columns to event context and background knowledge retrieved from web corpora, extracting features independently in each column before fusing them to combine local contextual information with external knowledge for judging causal relations between events. Ponti and Korhonen [11] applied feedforward networks with event-semantic features to discourse-level implicit causal identification, where event-semantic features included properties such as semantic roles, tense, and aspect of events, providing the model with event-structural information beyond surface-level lexical cues.

With advances in sequence modelling, bidirectional LSTMs (BiLSTMs) [90] became widely adopted for their ability to capture both forward and backward context. Dasgupta et al. [9] employed a linguistically informed BiLSTM for multi-word causal spans, feeding part-of-speech tags, dependency relations, and other linguistic features alongside word embeddings into the

BiLSTM to enhance the model’s awareness of causal expression structures. Li et al. [3] augmented BiLSTM-CRF with self-attention and transferred Flair contextual embeddings, and designed a dedicated tagging scheme to extract causal triplets directly rather than separately identifying entities and then classifying relations, thereby reducing error accumulation across the pipeline.

The introduction of pretrained transformer models brought further gains. Kadowaki et al. [8] fine-tuned BERT for event causality recognition while modelling annotator disagreement during training and testing multiple background knowledge injection strategies to explore how external knowledge can help the model better understand causal connections between events. Kruengkrai et al. [10] explored BERT variants with masked event spans, where the specific expressions of events are masked in the input to force the model to attend to structural causal cues between events rather than relying on the lexical features of the events themselves.

On the generalisation front, researchers began incorporating external knowledge and self-supervised learning to enhance cross-domain causal reasoning. Liu et al. [4] combined commonsense knowledge from ConceptNet [91] with a mention masking mechanism. ConceptNet is a large-scale open-source commonsense knowledge graph developed by the MIT Media Lab, organising knowledge as (head, relation, tail) triples with over 40 commonsense relation types (such as IsA, Causes, UsedFor, and HasProperty), encompassing approximately 8 million nodes and 21 million edges, and is designed to capture everyday human commonsense knowledge, widely used in tasks such as commonsense reasoning, knowledge graph completion, and causality mining. Liu et al. leveraged causal commonsense knowledge from ConceptNet as an external knowledge source while masking event mentions to encourage the model to learn event-agnostic causal patterns, thereby improving cross-scenario causal identification. Zuo et al. [5] induced transferable causal representations via self-supervised contrastive learning on external causal statements, pulling semantically similar causal pairs closer and pushing non-causal pairs apart so that the model can maintain robust causal identification even without target-domain annotated data. Cao et al. [6] integrated retrieved and generated descriptive knowledge with latent graph structure induction through structured attention, dynamically constructing implicit graph relations between events to assist causal judgement.

Although these methods have steadily improved causal extraction performance, their increasingly complex multi-component architectures also reflect the inherent limitation of the traditional paradigm: each performance gain tends to be accompanied by the introduction of additional external components or auxiliary modules, increasing system complexity and maintenance costs.

4.1.2. Generative Approaches

With the advancement of large language models (LLMs), generative approaches have begun to reformulate causal relation extraction as text generation tasks, reducing the dependence on

task-specific architectures and large annotated datasets [92]. Existing work in this direction falls into two categories: (1) task-dependent generative formulations, which apply LLMs to specific sub-tasks within the causal extraction pipeline, such as detection, classification, or phrase extraction; and (2) end-to-end structured generation approaches, which use a single model to directly generate structured triples from text.

Task-dependent Generative Formulations Although LLMs have been introduced into causal relation extraction, existing methods largely confine their use to a specific stage within the extraction pipeline rather than achieving unified causal triple generation. In the identification stage, KnowQA [24] reformulates document-level event causal relation extraction as binary question answering. It uses external IE tools (including the event detection model CLEVE, the argument extraction model BART-Gen, and the joint entity-relation extraction model JEREX) to automatically construct document-level event structures, which are then injected as structured context into LLM prompts to guide the model in answering whether a causal relation exists between events. In this framework, the LLM essentially serves as a pipeline-end classifier, and the quality of causal judgement depends on the upstream IE tools. In the detection and extraction stages, Keshavarz and Pourreza [20] designed a causal-connective-based Pattern RAG retrieval strategy that identifies the causal connective in the input sentence, retrieves structurally similar sentences from a pre-built causal pattern database, and dynamically injects them into the prompt as few-shot examples to assist the LLM in sentence-level causal detection and phrase labelling. GraphRAG-Causal [21] stores manually annotated causal graphs in a Neo4j graph database and retrieves relevant examples through a hybrid strategy combining vector embedding similarity with graph-structural cues to support LLM-based causal classification. CausalRAG [22] introduces causal path discovery at the retrieval level, first constructing a text graph from documents and retrieving query-relevant nodes, then expanding along graph paths and using an LLM to identify causal paths, thereby improving the causal relevance of the context provided to the generative model.

In domain-specific settings, Gopalakrishnan et al. [25] compared multiple LLMs (GPT-4, LLaMA2) against BERT-family models (BioBERT, DistilBERT, BERT) for causal span annotation in medical clinical practice guidelines, finding that fine-tuned BioBERT was more stable on small-scale medical datasets and revealing the inconsistency of LLMs in domain-specific causal extraction. However, the task itself remains token-level classification rather than generative causal extraction. FinCaKG and FinCaKG-Onto [93, 23] are among the most representative works in domain-specific causal knowledge graph construction. Targeting the financial domain, Xu et al. [93] first proposed FinCaKG, an end-to-end framework that constructs financial causal knowledge graphs from XBRL financial reports through a five-step pipeline: causal sentence detection, cause-effect span identification, financial keyword prediction, causal dependency representation, and graph generation. The framework employs BERT-based classifiers

for sentence detection, a graph-embedded model (ilab-FinCau) combining BERT embeddings with GNN+Bi-LSTM and Viterbi decoding for span identification, and connects extracted keyword pairs into causal chains where shared keywords across sentences serve as mediators linking different causal pairs together. While FinCaKG demonstrated the feasibility of automatic financial causal graph construction with strong domain coverage, its nodes were still represented as raw financial keyword strings without canonicalisation, which limited cross-graph interoperability and knowledge reuse. FinCaKG-Onto [23] subsequently extended this framework by introducing ontology alignment: in the entity linking stage, two strategies (exact string matching and generative entity linking via GENRE/mGENRE) are adopted to normalise extracted causal mentions to Wikidata identifiers; in the graph organisation stage, the FIBO financial ontology is used to organise the graph into a concept (T-Box) and instance (A-Box) hierarchy, with the `fincakg:cause` relation type and `rdfs:subClassOf` hierarchical relations establishing a taxonomic navigation system for causal knowledge. The core contribution of FinCaKG-Onto lies in resolving the node redundancy and ambiguity caused by raw-string node representations through entity linking and ontology alignment, enabling causal path reasoning and knowledge reuse. However, the framework remains a typical multi-stage pipeline at the extraction level, with each stage using a different model and errors potentially accumulating across causal sentence classification, span extraction, entity linking, and causality bonding. Furthermore, its causal extraction is restricted to intra-sentence causality, cross-sentence causal relations are not handled, and the ontology integration scheme is highly dependent on domain-specific expert resources (FIBO), requiring corresponding domain ontologies for transfer to other domains.

Overall, while these methods demonstrate the potential of LLMs for individual causal sub-tasks, the use of generative models remains confined to specific pipeline stages, and a unified framework for directly generating structured causal triples from raw text has yet to emerge.

End-to-end Structured Generation Approaches A separate line of research formulates information extraction as end-to-end text-to-structure generation, using a single model to directly produce structured outputs and thereby avoiding error propagation across pipeline stages. REBEL [85] reformulated relation extraction as autoregressive generation with BART-large, serialising triples with special tokens into linearised output and pretraining on large-scale silver data constructed from Wikipedia and Wikidata, with a natural language inference (NLI) model used to filter noisy distant supervision labels, demonstrating that a simple seq2seq formulation can generalise across datasets. Building on this paradigm, KnowGL [92] extended the generative formulation toward knowledge graph population by jointly generating entity mentions, canonical entity labels, entity types, and relation labels via BART-large in a single pass, and mapping the generated labels to Wikidata identifiers through label-to-ID lookup tables. While KnowGL demonstrated a more complete end-to-end pipeline from text to KG-compliant facts, it was designed for general-purpose knowledge graph population rather than causal ex-

traction. DEEPSTRUCT [30] unified multiple structure prediction tasks (such as named entity recognition and relation extraction) as triple generation through structure-aware pretraining on large-scale text–triple aligned corpora (e.g., T-Rex [94], ConceptNet [91], WebNLG [95]), using task prefixes to control the generation target and enabling cross-task transfer and zero-shot structure prediction. UIE [31] went further by unifying entity, relation, event, and sentiment extraction into a single text-to-structure framework, designing Structured Extraction Language (SEL) to provide a unified structured output representation (including Spot and Associate operations) and Structural Schema Instructor (SSI) to encode task schemas as part of the input to control extraction behaviour, achieving cross-task transfer across multiple IE tasks based on a T5 encoder-decoder architecture. GoLLIE [29] addressed the zero-shot setting by treating annotation guidelines as first-class inputs, representing IE schemas in Python code style (class definitions with docstrings as annotation guidelines), and training an LLM to follow schema descriptions for extraction, with regularisation strategies including class shuffling, class dropout, guideline paraphrasing, and label masking to enhance generalisation, demonstrating strong cross-domain generalisation to unseen schemas. Addressing the faithfulness challenge, CGT [32] introduced triplet-level contrastive learning, using correct triples as positive samples and constructing negative samples through random entity replacement, training the model to explicitly penalise outputs unsupported by the input text while generating correct triples. These methods have achieved end-to-end generation for general structural extraction, but have not yet been adapted to the semantic specificity of causal relations.

When extended to specific domains, such frameworks typically require domain-specific fine-tuning and customised design. KGLM [26] fine-tuned ChatGLM with a four-component structured prompt template comprising system role definition, triple schema examples, output format constraints, and chain-of-thought reasoning guidance, to directly generate knowledge triples from unstructured lung cancer texts. Dagdelen et al. [28] adopted a similar paradigm for materials science, fine-tuning GPT-3 and Llama-2 to map text passages directly to user-defined structured schemas (such as JSON objects), enabling joint named entity recognition and relation extraction in a single generation step. Their approach handles hierarchical, multi-entity scientific relations that are difficult to capture with pairwise extraction methods, and demonstrates that a few hundred task-specific annotations can be sufficient for effective fine-tuning. Yan [27] extended domain-specific generative extraction to multi-causal event causality mining in financial texts, combining a fine-tuned LLaMA backbone with graph-based multi-hop reasoning over structured event tuples to detect both direct and indirect causal chains, addressing the limitation of prior methods that typically handle only single-cause, sentence-level, explicitly marked relations. While these domain-specific methods achieve end-to-end or near-end-to-end generation within their respective domains, they share a common limitation: high dependence on domain-specific fine-tuning data, customised schemas or prompts, and in some cases auxiliary reasoning modules, resulting in limited cross-domain transferability.

This reveals a fundamental trade-off that remains unresolved in existing end-to-end methods: general structural extraction frameworks (e.g., UIE [31], DEEPSTRUCT [30], GoLLIE [29]) possess cross-task generalisation but have not been optimised for causal semantics; domain-specific methods (e.g., KGLM [26]) effectively extract structured knowledge but lack generalisability. How to simultaneously address causal specificity, cross-domain robustness, and output faithfulness within an end-to-end generative framework remains an open research question.

4.2. Causal Knowledge Graph Construction

4.2.1. Construction from Predefined Resources

Causality is most directly represented as pairwise text, taking the form of word pairs (e.g., investment–bankruptcy) or span pairs. Many early and existing works construct causal knowledge graphs from such causal pairs or predefined resources. Based on the core strategies employed, existing methods can be broadly organised into three directions: large-scale pattern-based causal pair accumulation, domain-specific causal graph construction, and node canonicalisation through ontology alignment and semantic clustering.

Large-scale Pattern-based Causal Pair Accumulation CauseNet [1] is one of the most representative early works in this direction. The system bootstrapped causal word pairs from ClueWeb12 (approximately 733 million web pages) and Wikipedia at scale: starting from seed causal pairs (e.g., smoking → cancer), it learned linguistic patterns (e.g., “lead to,” “cause,” “result in”), then applied dependency parsing and BiLSTM-CRF for concept extraction, ultimately constructing an open-domain causality graph containing approximately 11.6 million mayCause relations with provenance information retained for each relation. CauseNet demonstrated the feasibility of automatically accumulating causal knowledge from web-scale data, but the graph used only a single relation type and represented nodes as raw noun phrases, resulting in a sparse and unnormalised graph. ASER [39] extended this line of work from the perspective of event-level commonsense knowledge, using 18 hand-designed dependency patterns to extract eventualities (i.e., activities, states, and events represented as dependency-graph-like structures) from over 11 billion tokens, and labelling inter-eventuality edges with 15 PDTB-style discourse relations including causal types such as Reason, Result, and Condition. The resulting graph contains approximately 439 million eventuality nodes and 649 million edges. A key extension in ASER is the introduction of a Probase-based conceptualisation layer that abstracts concrete eventualities into conceptualised eventualities, thereby mitigating node redundancy and graph explosion without relying on external knowledge base linking. Hassanzadeh et al. [42] similarly targeted large-scale causal knowledge mining, applying weakly supervised methods to extract cause–effect phrase pairs from approximately one billion news sentences. The system used word2vec and BERT embeddings for similarity-based clustering of surface variants (e.g., group-

ing “global warming” with “climate change”) and exposed the extracted knowledge through APIs for causal search, evidence retrieval, and domain-specific causal mind-map construction, positioning causal knowledge mining as an operational system service rather than solely an academic benchmark task. In earlier work, Radinsky et al. [44] proposed the Pundit system, which mined cause–effect event pairs from 150 years of news headlines and applied ontology-based abstraction to generalise specific events into an abstraction tree for forward causal reasoning (i.e., predicting plausible future effect events from a given cause event). While not a causal KG construction method per se, this work demonstrated how structured event representations and ontology-level generalisation can support causal prediction from textual knowledge.

Domain-specific Causal Graph Construction In domain-specific settings, finance and biomedicine represent the two most active directions. In the financial domain, Sakaji and Izumi [37] proposed a bilingual causality extraction model that combines BERT encoding, graph attention networks (GAT) over universal dependency trees, and a clue-expression encoder (which identifies specific causal trigger phrases such as “because,” “due to,” and their Japanese equivalents) to extract nested causalities (i.e., sentences containing multiple layers of causal structure) from English and Japanese financial texts. Izumi and Sakaji [43] further linked extracted causal pairs into causal chains using word-embedding cosine similarity between one pair’s effect and the next pair’s cause, and built a prototype interactive system that allows users to trace upstream causes or downstream consequences from a given event. In the business domain, Text2CausalGraph [36] proposed a framework for constructing causal knowledge graphs from S&P financial company 10-K reports. Its distinctive feature is the integration of causal extraction with a stakeholder taxonomy, which classifies the cause and effect in each causal relation according to the stakeholder group affected (e.g., investors, customers, employees, society), so that the resulting causal knowledge reflects not only “what causes what” but also “who is affected,” supporting decision-oriented business analysis.

In the biomedical domain, Lyu et al. [38] constructed a multisource causal knowledge graph for clinical decision support of diabetic nephropathy, integrating causal triples from three sources: SemMedDB (literature mining), UpToDate (clinical guidelines), and Churchill’s Pocketbook of Differential Diagnosis (textbook). Entity alignment was performed using UMLS concept unique identifiers (CUIs), and two noise-reduction pruning strategies based on co-occurrence ratio and causality ratio were designed to reduce knowledge noise from literature mining. A key contribution of this work is that it mapped the extracted causal knowledge to real-world electronic health record (EHR) data and evaluated graph quality through an actual diagnostic reasoning task, rather than relying solely on intrinsic evaluation. Domingo-Fernández et al. [41] manually curated a mechanistic cause-and-effect knowledge graph for COVID-19 pathophysiology from 160 research articles, encoding the knowledge in Biological Expression Language (BEL) with explicit directed causal predicates (e.g., increases, decreases, causesNoChange). Nodes

were canonicalised through standard biomedical namespaces including HGNC, ChEBI, GO, and MeSH. This work demonstrated the value of high-precision manual curation for building mechanistic causal networks, though scalability remains limited by curation cost.

Ontology Alignment, Semantic Clustering, and Evaluation On the node canonicalisation and ontology alignment front, FinCaKG-Onto [23] is the most representative work. The framework linked causal nodes to Wikidata identifiers through entity linking and organised the graph into a concept (T-Box) and instance (A-Box) hierarchy via the FIBO financial ontology, effectively resolving the node redundancy and ambiguity caused by raw-string node representations in earlier causal graphs and enabling taxonomic navigation and causal path reasoning. From the perspective of evaluation infrastructure, Hassanzadeh [33] proposed WikiCausal, a corpus and evaluation framework for end-to-end causal knowledge graph construction that uses existing Wikidata causal relations as a recall reference and LLM-assisted verification for precision estimation, providing a reusable benchmark for systematically comparing causal KG construction methods. In a related earlier effort, Hassanzadeh [40] attempted to construct an event–consequence knowledge graph from Wikidata and Wikipedia by first extracting existing causal relations from Wikidata as a base graph and then augmenting it with new causal edges mined from Wikipedia text through an unsupervised QA-based extraction pipeline. This work framed causal KG construction as a joint knowledge engineering and text mining problem, but also exposed practical difficulties in event entity linking and relation directionality judgement.

In terms of semantic clustering, Tan and Paul [35] observed that directly extracted causal graphs tend to be sparse and poorly connected, and proposed applying SimCSE-based semantic clustering after extraction to compress scattered causal argument nodes into clustered topics, transforming the graph from multiple isolated components into a fully connected structure and significantly improving interpretability and downstream usability, though cluster labels remain non-canonical concept identifiers. From a causal chain quality perspective, CK-CEVAE [34] demonstrated that valid causal pairs do not necessarily form valid causal chains, proposing a multi-stage variational architecture to model the semantic continuity of causal chains and introducing domain consistency features to detect where causal chains break.

Overall, construction from predefined resources has evolved from early word-pair accumulation to a multi-faceted paradigm incorporating semantic clustering, domain taxonomies, ontology alignment, and multisource knowledge integration. FinCaKG-Onto has mitigated node redundancy through ontology integration, Lyu et al. have advanced graph quality control through multisource integration and noise pruning, and WikiCausal has provided evaluation infrastructure for systematic comparison. However, ontology alignment solutions remain highly dependent on domain-specific expert resources with limited generalisability, while methods without ontological support still face non-standardised node representation and difficult knowledge

reuse. Moreover, most existing methods are designed for specific domains (finance, medicine, news), and a unified cross-domain approach to causal knowledge graph construction remains to be explored.

4.2.2. Construction with Generative Models

Beyond causal relation extraction itself (Section 3.1.2), generative models have also been applied to the graph construction and organisation stage, that is, how to transform extracted knowledge into structured, reusable graphs. Based on their focus, existing works can be examined from three perspectives: the ability of generative paradigms to model graph structures, automatic schema and ontology induction, and graph quality verification and refinement.

Generative Paradigms for Graph Structures KG-S2S [57] unified different types of knowledge graph completion tasks (static, temporal, and few-shot) as T5-based seq2seq generation, converting KG facts into flat text and using constrained decoding (i.e., restricting generated tokens to valid entities already present in the KG) to produce target entities, thereby demonstrating the viability of generative paradigms for graph-structured tasks. EDC [47] extended the generative paradigm from graph completion to graph construction, proposing an extract–define–canonicalize pipeline in which an LLM first performs open triplet extraction, then generates natural-language definitions for the extracted schema elements, and finally canonicalises them via embedding similarity and LLM verification, enabling the framework to operate with or without a predefined schema. Graphusion [50] further advanced generative graph construction from local extraction toward global fusion: after extracting candidate triples via an LLM, it introduces a KG Fusion module that takes a global view to merge semantically similar entities, resolve conflicting relations, and infer novel triples not present in the local extractions. This progression from local extraction to global fusion reflects a broader evolution in generative graph construction, from how to generate triples toward how to organise a coherent graph. FinDKG [49] extended this line into temporal graph evolution, combining LLM-based triplet extraction (fine-tuning Mistral-7B on GPT-4-generated training data from financial news) with a type-aware temporal graph learning model (KGTransformer) for dynamic link prediction, demonstrating that generative construction can be coupled with graph-level temporal learning within a single domain-oriented pipeline.

Schema and Ontology Induction Graph construction involves not only fact extraction but also schema and ontology design. Traditional methods typically rely on manually predefined schemas, but recent work has begun exploring automatic schema induction using LLMs. Trajanoska et al. [56] provided early evidence that ChatGPT can directly generate OWL [65] ontologies and RDF [64] instances through detailed prompting, demonstrating LLM capability for ontology engineering and KG population, though the generated outputs lacked cross-document

consistency and often required manual correction. AutoSchemaKG [46] pushed this idea to a much larger scale, proposing a fully autonomous framework that jointly extracts triples and induces schemas through conceptualisation (i.e., abstracting concrete entities and events into semantic categories), modelling both entities and events as first-class nodes and constructing knowledge graphs with over 900 million nodes and 5.9 billion edges without any predefined ontology. TKGCon [52] approached the problem from a different angle, targeting theme-specific fine-grained graph construction: it first builds an entity ontology from Wikipedia categories, then uses an LLM to generate a candidate relation ontology for category pairs, and subsequently performs entity recognition and relation extraction under ontology guidance, demonstrating how ontology-guided decomposition can make LLM-based graph construction more precise and thematically coherent. Kommineni et al. [53] embedded LLMs within a more traditional ontology engineering workflow, driving ontology design from competency questions (i.e., predefined questions that the ontology and KG should be able to answer), then using RAG to answer these questions and populate the KG, preserving the requirements-driven ontology design rationale that many end-to-end approaches bypass. Hofer et al. [54] explored a different mode of LLM–graph construction integration: rather than having the LLM generate triples directly, they used LLMs to generate RML mapping files (i.e., declarative transformation rules that convert semi-structured data into RDF triples), which are then executed by standard mapping engines. This approach positions the LLM as a self-configuration agent for graph construction pipelines rather than a replacement for the entire construction workflow.

Graph Quality, Verification, and Refinement As the scale of LLM-generated triples grows, ensuring graph quality becomes a critical concern. iText2KG [48] addressed this from the construction process itself, designing a zero-shot incremental framework that performs entity and relation deduplication during extraction rather than deferring it to post-processing, thereby reducing semantic duplication at the source. SAC-KG [51] proposed an iterative domain KG construction framework comprising three modules: a Generator that produces triples from domain corpora through retrieval-augmented prompting, a Verifier that detects and corrects generation errors using rule-based checks (covering error types such as quantity insufficiency, format errors, head entity errors, and head–tail contradictions), and a Pruner that uses a T5 binary classifier to decide whether each tail entity should continue expanding into the next graph level, transforming graph construction into a controlled iterative expansion process. KGLM [26] similarly demonstrated domain-specific iterative construction by fine-tuning ChatGLM with structured prompt templates to directly generate knowledge triples from lung cancer texts, though the approach remains highly dependent on domain-specific fine-tuning data and customised prompts. On the verification and evaluation front, KGValidator [55] proposed an LLM-based triple validation framework operating under the open-world assumption (i.e., triples absent from the KG should not automatically be treated as incorrect), combining the LLM’s parametric knowledge

with evidence retrieved from Wikidata and web search to verify candidate triples. LLMKG+ [45] further focused on triple-level semantic redundancy: its hierarchical expansion filter jointly checks whether a generated triple is factually correct and whether it is semantically novel (i.e., not equivalent in meaning to an existing triple), and introduces the KG Reconstruction Test, which computes precision and recall while accounting for semantic equivalence rather than relying solely on exact matching.

Despite the notable progress achieved across generative paradigms, schema induction, and quality control, these methods still share several common limitations. The construction process mostly remains sequential, essentially embedding LLMs into individual pipeline stages rather than achieving truly end-to-end graph generation. Systematic quality evaluation is lacking, especially with regard to causality-specific metrics. Most methods do not integrate existing ontological resources, limiting knowledge reuse and cross-graph navigation. Finally, these frameworks are generally designed for general-purpose knowledge graph construction and are not tailored to causal relations, leaving the proportion and quality of causal knowledge in the resulting graphs unassured.

5. Problem Statement and Research Questions

5.1. Problem Statement

The literature reviewed above reveals a central gap in current research: while individual methods exist for causal relation extraction and causal knowledge graph construction, existing frameworks remain limited in unifying these two processes into a single generative workflow that can reliably produce structured causal knowledge from multi-domain text sources.

On the extraction side, traditional approaches are fundamentally constrained by their dependence on manual annotation, domain-specific design, and error propagation across multi-stage pipelines. Generative methods have improved flexibility, but as analysed in Section 3.1.2, a fundamental trade-off persists: task-dependent methods confine LLMs to individual pipeline stages rather than generating structured causal outputs directly; general-purpose end-to-end IE frameworks (e.g., DEEPSTRUCT [30], UIE [31], GoLLIE [29]) offer cross-task generalisation but have not been optimised for the semantic specificity of causal relations; and domain-specific methods (e.g., KGLM [26]) achieve effective extraction but are tightly bound to fine-tuning data and customised prompts, limiting transferability. How to simultaneously address causal specificity, cross-domain robustness, and output faithfulness within a unified generative framework therefore remains an open problem.

At the graph construction level, the limitations are equally evident. As discussed in Section 3.2.1, resource-based methods have evolved from word-pair accumulation to incorporating clustering, taxonomies, and ontology alignment, yet methods lacking ontological support still face

non-standardised node representation and difficult knowledge reuse, while ontology-dependent solutions have limited generalisability. As discussed in Section 3.2.2, generative models have been applied to graph construction, but the process mostly remains sequential, systematic quality evaluation is lacking, existing ontologies are rarely integrated, and the frameworks are generally not tailored to causal relations. Consequently, the central challenge is no longer whether local causal information can be extracted, but how causal extraction and graph construction can be integrated into a coherent framework for reliable structured causal knowledge generation.

5.2. Research Aim

This study aims to develop an LLM-based generative framework that can directly produce structured causal triples from multi-domain text sources and, through necessary but lightweight post-generation processing, organise them into coherent, verifiable, and extensible causal knowledge graphs. Rather than eliminating all sequential steps, the framework seeks to concentrate the core extraction process within a single generative model, while retaining graph-oriented refinement steps such as normalisation, entity linking, ontology alignment, and graph organisation.

5.3. Research Questions

5.3.1. Main RQ

How can an LLM-based generative framework be designed to directly produce structured causal triples from multi-domain text and organise them into coherent, lightweight causal knowledge graphs?

5.3.2. RQ1: Framework Design

How can an LLM-based generative model be designed to jointly produce structured causal triples that capture causal entities, relations, and span boundaries in a single generation pass?

5.3.3. RQ2: Reliability and Hallucination

What types of hallucination arise in LLM-based causal triple generation, and how can they be mitigated to improve faithfulness and consistency within the framework?

5.3.4. RQ3: Evaluation

How should generated causal triples and causal knowledge graphs be evaluated in terms of extraction accuracy, graph completeness, faithfulness to source text, and structural consistency?

6. Methodology

6.1. Overall Framework

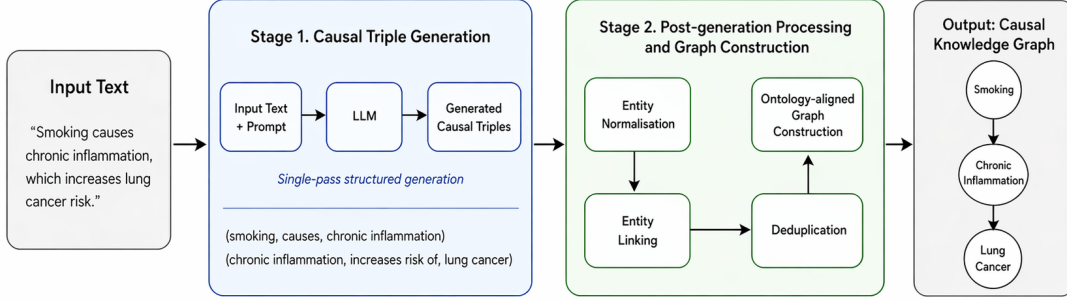


Figure 4: Overall framework of the proposed method. The framework consists of two stages: causal triple generation from input text and post-generation processing for ontology-aligned causal knowledge graph construction.

As illustrated in Figure 4, the proposed framework consists of two stages that together form a complete pipeline from raw text to a causal knowledge graph. In the first stage, Causal Triple Generation, an LLM produces structured causal triples from the input text in a single generation pass. In the second stage, Post-generation Processing and Graph Construction, the generated triples undergo entity normalisation, entity linking, and deduplication, and are then transformed into a representation aligned with a domain-independent causal ontology (detailed in Section 5.2) for graph construction. The core design principle is to concentrate the main extraction workload within the generative model itself, while the subsequent processing stages convert raw triples into a well-structured and reusable causal knowledge graph. The resulting outputs are subsequently assessed through both triple-level and graph-level evaluation, with particular attention to hallucination and faithfulness.

6.2. Task Formulation

The task is defined as follows: given a multi-domain text passage, the model directly outputs structured causal triples in a single generation pass. The output takes the form of structured Python code following the GoLLIE [29] paradigm, where each triple is represented as an instantiated Python class object that can be directly parsed into structured data. A single passage may yield zero or more triples. This formulation fundamentally differs from prior causal detection, classification, or QA-style extraction in that the model jointly completes causal entity identification, relation determination, and span boundary capture within one generation step.

The output schema is grounded in the causal ontology proposed by Jaimini et al. [58], which offers two properties that make it particularly suitable for this study. First, it introduces the

Mediator as a first-class element. A mediator is an intermediate event that lies between a cause (treatment) and an effect (outcome) in a causal chain, transmitting or influencing the causal relationship between them. For example, in the sequence “it is raining $\rightarrow$ pavement is wet $\rightarrow$ pavement is slippery,” the event “pavement is wet” serves as the mediator. A causal relation may involve zero or more mediators. By explicitly distinguishing direct causation (causes) from mediated causation (causesWith), the ontology enables the representation of causal pathways where a cause leads to an effect through one or more intermediate mechanisms, allowing the output to go beyond flat cause–effect pairs and capture richer causal structures including indirect causation and multi-hop transmission effects. Second, it is designed as a domain-independent Ontology Design Pattern (ODP) that can be lightweight extended to any existing domain ontology, maintaining generality while minimising the complexity of practical deployment. These characteristics ensure that the same output schema can be applied across domains without redesign.

6.3. Causal Triple Generation

6.3.1. Guideline-Constrained Structured Generation

The generation stage adopts the annotation-guideline paradigm proposed by GoLLIE [29]. Specifically, the definition of causality, span boundary rules, and the output schema are encoded as Python class definitions with docstrings (i.e., structured documentation strings embedded within the class definition that describe the attributes and constraints of each class) and provided to the model as first-class inputs. These guidelines specify what counts as a causal relation, how causality is distinguished from correlation, whether implicit causality is included, how cause and effect spans are delimited, and what structured format each output triple should follow. GoLLIE adopts Python code as its output format because code language models exhibit stronger structural reasoning over class hierarchies and typed attributes, which naturally enforces output consistency and makes the results directly parseable by downstream systems without additional format conversion. The model output thus consists of instantiated Python objects, each corresponding to a causal triple. Note that the Python code format is adopted following GoLLIE’s original design; depending on experimental findings, the output format may be adjusted to direct triple generation in later stages of the study.

Given the current hardware constraints, the largest feasible model size for this study is around 14B parameters. Accordingly, the experimental models are restricted to the 13B–14B range. Within this range, one model specifically designed for guideline-following information extraction is adopted as the primary baseline, together with two strong general-purpose 14B models as comparative baselines, in order to assess whether the guideline-constrained mechanism itself provides additional benefits for causal extraction.

HiTZ/GoLLIE-13B HiTZ/GoLLIE-13B is a 13B model fine-tuned from Code Llama 2 by the HiTZ Basque Center for Language Technology for zero-shot information extraction. Its key feature is the ability to follow detailed user-provided annotation guidelines at inference time, thereby producing structured outputs for dynamically defined tasks such as named entity recognition, event extraction, and relation extraction. It is therefore used as the primary guideline-constrained baseline for evaluating the effect of explicit annotation guidelines on causal triple generation [29].

wanlige/li-14b-v0.4 wanlige/li-14b-v0.4 is a 14.8B model based on Qwen2.5-14B-Instruct and further enhanced through model merging [96]. According to its Hugging Face model card, it is ranked first among models with up to 15B parameters on the Open LLM Leaderboard. By integrating capabilities from models strong in coding, mathematical reasoning, and text generation, it achieves strong overall performance in general instruction following and complex reasoning. In this study, it is included as a strong merged-model baseline for comparison with guideline-constrained extraction.

JungZoona/T3Q-qwen2.5-14b-v1.0-e3 JungZoona/T3Q-qwen2.5-14b-v1.0-e3 is a LoRA-based post-trained model built on Qwen2.5-14B-Instruct, with approximately 14.8B parameters [97]. According to its Hugging Face model card, the model also ranks highly on the Open LLM Leaderboard, including strong performance among models below 32B parameters. It is reported to perform strongly in instruction following, long-chain reasoning, and dialogue consistency, and is included as a representative high-performing instruction-tuned general-purpose model. This allows us to examine how far a strong general backbone can support causal structured extraction without explicit guideline-constrained training.

The rationale for this model configuration is twofold. First, it keeps model scale roughly controlled, reducing the extent to which performance differences are driven mainly by parameter size. Second, it enables a more direct comparison between explicit guideline-constrained generation and strong general-purpose models of similar scale, particularly in terms of faithfulness, structural stability, and causal extraction quality.

6.3.2. Pattern-based Retrieval-Augmented Generation

While GoLLIE possesses zero-shot capability, the semantic complexity of causal relations means that pure zero-shot generation may not guarantee sufficient extraction quality. To address this, the study draws on the Pattern RAG approach proposed by Keshavarz and Pourreza [20], attaching a causal pattern database to the generation pipeline. This database is derived from causal patterns extracted from three datasets (SemEval-2010 Task 8 [98], ADE [2], and the extended dataset by Li et al. [3]). At inference time, a retrieval algorithm selects causal patterns

structurally similar to the input text and injects them as few-shot examples into the prompt, extending GoLLIE’s zero-shot generation into RAG-enhanced few-shot generation. The original method uses kNN for pattern retrieval; given the limitations acknowledged by its authors, this study will explore improved retrieval strategies.

6.4. Post-generation Processing and Graph Construction

The structured causal triples produced by the generation stage require further processing before they can be organised into a reusable causal knowledge graph. First, the entities within the triples are normalised: surface forms are unified, semantically duplicate nodes are merged, and conflicting triples are detected and resolved, thereby eliminating node redundancy caused by variant expressions of the same concept. The normalised entities are then linked to external knowledge bases such as Wikipedia [99] to enhance interpretability and cross-graph interoperability. Finally, the structured Python output is converted into a semantic representation expressed in RDFS/OWL and aligned with the causal ontology of Jaimini et al. [58], ensuring ontology-level consistency and reusability. Upon completion of these steps, the causal knowledge graph is constructed.

6.5. Hallucination Analysis and Mitigation

LLM-based causal triple generation inevitably faces the risk of hallucination, which directly affects output reliability (corresponding to RQ2). This study will pre-define several hallucination types, including but not limited to: causal relations unsupported by the source text, incorrect span boundaries, erroneous entity links, and relation types inconsistent with the ontology. On the mitigation side, the annotation guideline constraints form a first line of defence by narrowing the generation space; the causal patterns retrieved via RAG provide factual grounding for the generation; and the post-generation normalisation, linking, and ontology alignment steps further filter inconsistent outputs. The specific implementation details and effectiveness of each strategy will be progressively verified and adjusted through experimentation.

6.6. Evaluation Design

6.6.1. Datasets

To evaluate the applicability and robustness of the proposed framework under different data conditions, this study selects three representative datasets. These datasets differ substantially in domain source, annotation granularity, data scale, and noise level, but all provide a basis for causal knowledge extraction.

The extended dataset by Li et al. The extended dataset by Li et al. [3] is a re-annotated and expanded version of SemEval-2010 Task 8. It adopts a dedicated causality tagging scheme that

formulates causal extraction as a fine-grained sequence labeling task, with explicit annotation of cause spans and effect spans, including multi-word phrases. Unlike the original SemEval dataset, in which Cause–Effect is only one relation label among several semantic relation types, this extended version is more suitable for evaluating fine-grained causal span extraction and structured causal representation. It is therefore useful for testing model performance in sentence-level causal triple generation, especially in cases involving multi-word expressions and complex boundary identification.

CauseNet CauseNet [1] is one of the largest automatically constructed causal knowledge resources currently available. It was built by Heindorf et al. through large-scale extraction from web data and contains approximately 11.6 million mayCause relations. Its nodes are represented as raw text strings, and each edge is associated with provenance information. Compared with manually annotated datasets, CauseNet has the advantage of very broad coverage and is particularly useful for large-scale causal knowledge discovery, causal graph construction, and open-domain causal representation learning. However, because it is constructed entirely through unsupervised pattern mining, it also contains substantial noise, redundancy, and semantic ambiguity. In this study, CauseNet is therefore treated as a large-scale open-domain causal resource for examining the framework’s behavior under weakly supervised and noisy conditions.

ADE The ADE dataset [2] is a classic corpus for adverse drug event relations, mainly derived from biomedical literature and case reports. It contains sentence-level annotations of causal relations between drugs and adverse effects, and is widely used as a benchmark for causal detection and causal extraction in the biomedical domain. Compared with open-domain data, ADE is more domain-specific, and its causal expressions rely more heavily on medical terminology and specialized context. It is therefore suitable for evaluating the transferability of the proposed framework to professional-domain text and its applicability to medical causal knowledge extraction.

Although these three datasets differ in source, scale, and annotation format, they will be preprocessed into a unified input–output format so that they can be used within the same generation and graph construction framework. Among them, the extended dataset by Li et al. provides relatively high-quality sentence-level annotations and is suitable for evaluating fine-grained causal extraction; ADE provides realistic causal instances in a specialized domain and is useful for assessing cross-domain transfer; and CauseNet provides a large-scale but noisier open-domain causal resource, making it suitable for testing the robustness of the framework under weakly supervised conditions. Evaluating the framework on these three resources together makes it possible to analyze its performance more comprehensively across different data conditions.

6.6.2. Triple-level Evaluation

At the triple level, accuracy, Precision, Recall, and F1 are used as core metrics to assess the quality of generated triples against the annotated ground truth in each dataset, which also serves as the basis for evaluating faithfulness to the source text.

6.6.3. Graph-level Evaluation

At the graph level, given the absence of standardised automatic evaluation methods, this study follows the approach of FinCaKG-Onto [23] by combining human evaluation with LLM-assisted judgement. Specifically, multi-hop causal chains (e.g., two-step causal paths) will be randomly sampled from the constructed graph, and their semantic coherence and factual correctness will be assessed by human annotators or LLM-based judges to evaluate structural consistency and overall graph quality.

6.6.4. Comparative Experiments

To systematically assess the contribution of each design component, the study plans the following comparisons: GoLLIE versus the two general-purpose 14B models (T3Q-qwen2.5-14b and li-14b), to evaluate whether explicit guideline-constrained training provides additional benefits over strong instruction-tuned backbones of comparable scale; and zero-shot versus RAG-enhanced generation, to evaluate the contribution of causal pattern retrieval to extraction quality. In addition, further datasets from domains such as finance and science may be incorporated in future work, with their results compared against the initial three datasets to further validate cross-domain robustness.

7. Research Timeline



Figure 5: Research timeline

The research is planned over a 17-week period from 13 April to 12 August 2026, organised into five phases. Phase 1 (3 weeks) establishes the foundation by finalising the causal annotation guideline and output schema, setting up the experimental environment for GoLLIE and the baseline model, and downloading and organising the three evaluation datasets along with exploring additional domain-specific datasets. Phase 2 (5 weeks) constitutes the core experimental work, implementing the guideline-constrained structured generation and retrieval-augmented few-shot mechanism, running extraction experiments across all datasets, and conducting the comparative experiments. Phase 3 (3 weeks) focuses on transforming the generated triples into a causal knowledge graph through entity normalisation, Wikipedia linking, and conversion into ontology-aligned semantic representations. Phase 4 (3 weeks) carries out the evaluation at both the triple and graph levels and analyses hallucination types and mitigation effectiveness. Phase 5 (3 weeks) is dedicated to thesis writing, result integration, and final submission.

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A. AI Usage Statement

Artificial intelligence tools were used only as auxiliary support during the preparation of this proposal, mainly for the following two purposes.

First, AI tools were used to assist with literature search and bibliographic checking. During this process, the author required the AI system to provide verifiable sources, which were used as search leads rather than as direct evidence. All relevant publications, publication information, and reference entries were subsequently checked by the author against the original papers, official publication pages, or database records before being used in the proposal.

Second, AI tools were used for language-level assistance, including grammar checking, wording refinement, and conversion of author-written content into LaTeX format. These uses were intended to improve the clarity, consistency, and formatting of the text, rather than to generate the research ideas or core content.

The overall structure, research design, main content, figures, and final arguments of this proposal were completed by the author. AI-generated suggestions were reviewed, selected, and revised by the author. The author takes full responsibility for the final content of this proposal.